

Bayesian Edge Detection

Lecture notes

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Outline

Edge detection as a decision problem

Filter response as observation

Learning the class-conditional distributions

From the likelihood ratio to a decision

Ambiguity and multiple cues

Semantic segmentation

Generative models and domain transfer

Summary

Recap: Bayes decision theory

- ▶ Bayes decision theory needs five things: an observation, two class-conditional distributions, a prior, a loss, and a rule that minimises the expected loss. The previous lecture developed all five in the abstract.
- ▶ For a vision problem the new question is what the *observation* should be when the data is an image.
- ▶ The answer is a *filter response*. Once the observation is fixed, the rest of the previous lecture carries over unchanged.



one number computed from the image

Figure 1: The structure of Bayes decision theory, written in the notation of this lecture. Only the first box is new.

Definition of an edge

- ▶ We treat as an *edge* both the boundary of an object and a significant change of texture, such as the lettering printed on a mission patch or on a t-shirt.
- ▶ Everything else is background, including strong intensity variation that lies *inside* a region: individual hair strands, fabric weave, foliage.
- ▶ This is a convention, and it is the one used by the annotated datasets we learn from. A different convention would give different class-conditional distributions, and so a different detector.

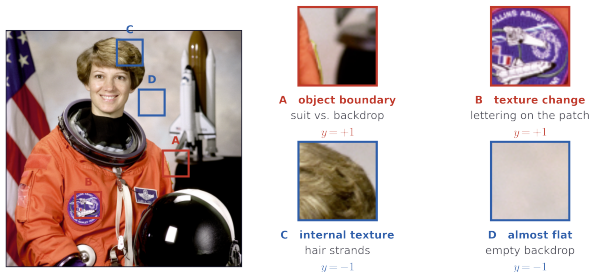


Figure 2: Four locations in one photograph. A and B carry the label $y = +1$, C and D carry $y = -1$. Photograph: NASA portrait of Eileen Collins, public domain.

Large gradients are usually edges

- ▶ The intensity changes rapidly across an object boundary, so the magnitude of the image gradient is an obvious cue, and a good one.
- ▶ Hair, foliage and fabric also produce gradients as large as those at a true boundary, and a detector that thresholds the gradient reports all of them as edges.
- ▶ The research question is: how to distinguish the edge from background in an image? The filter response is treated as evidence to be weighed rather than as a verdict.

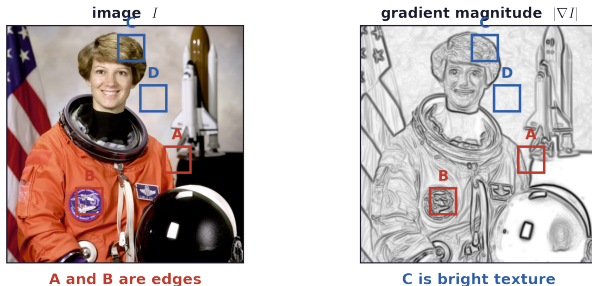


Figure 3: The gradient magnitude responds at the boundary A, at the lettering B, and just as strongly at the hair texture C.

Formulation of the edge detection problem

- ▶ At each pixel x the image has an intensity $I(x)$ and carries a variable $y(x) \in \{+1, -1\}$, where $+1$ indicates an edge and -1 indicates the opposite.
- ▶ Edges are defined *on* pixels, rather than on the boundaries between pixels. Every pixel receives exactly one label.
- ▶ The task is a binary decision, repeated independently at every pixel: given I : decide $y(x) \in \{+1, -1\}$ at every pixel x from $f(I(x))$.

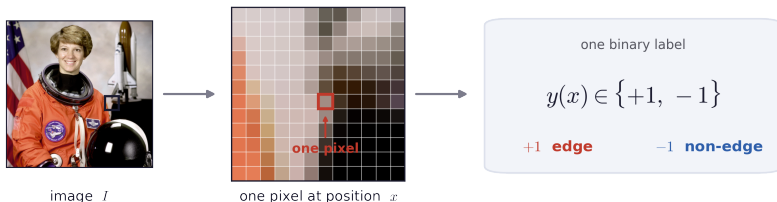


Figure 4: One pixel, one binary label. The zoom shows individual pixels, and the marked pixel is a cell rather than a corner.

From an image to a number

- ▶ Bayes decision theory needs a scalar measurement that can be compared against class-conditional distributions.
- ▶ We therefore apply a filter $f(\cdot)$ to the image in a small neighbourhood of x and keep the number it returns. That number, $f(I(x))$, is the observation.
- ▶ If the filter is tuned to edges, then $f(I(x))$ is likely to be higher when an edge is present than when it is absent. This is the property the whole method rests on, and the property we will quantify.

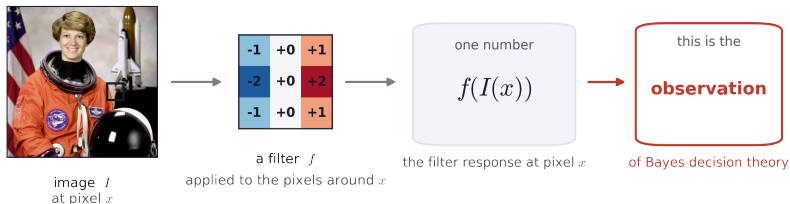


Figure 5: Image, filter, response. The response plays the role of the observation in Bayes decision theory.

Notation

x	the pixel position
$I(x)$	the image intensity at that pixel
$f(\cdot)$	a filter
$f(I(x))$	the filter response, written f for short
$f_i(I(x)), i = 1, \dots, M$	the responses of a filterbank
$y(x) = +1$	the pixel is an edge
$y(x) = -1$	the pixel is not an edge
T	the decision threshold
$\alpha(x), \alpha_T(x)$	the decision rule

Two meanings of x

The previous lecture wrote the generic observation as x . Here x is the pixel position and the observation is $f(I(x))$. When comparing the two lectures, substitute $f(I(x))$ for x throughout.

The simplest filter: the derivative

- ▶ Scan one row of pixels. Inside a region the intensity is roughly constant, so its derivative is near zero. Where the intensity steps from one level to another the derivative is large.
- ▶ This is the entire intuition behind derivative filters: they measure how fast the image is changing, and boundaries are where it changes fastest.
- ▶ In practice the derivative is computed after smoothing, because differentiation amplifies noise. The standard choice is the derivative of a Gaussian,

$$\frac{\partial}{\partial x}(G_\sigma * I) = \left(\frac{\partial G_\sigma}{\partial x}\right) * I, \quad G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2},$$

which is itself a filter. The scale σ selects the size of structure the filter responds to.

The simplest filter: the derivative

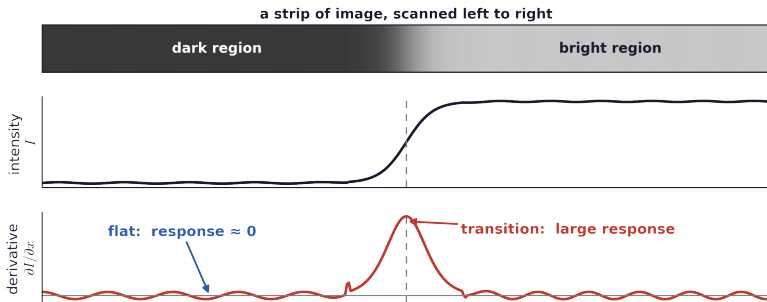


Figure 6: A strip of image scanned left to right, its intensity profile, and the derivative of that profile. Flat regions give a response near zero, and the transition gives a large response.

- The response is large over a short *band* around the transition, several pixels wide. Converting that band into a one-pixel-wide edge map is a separate step, usually non-maximum suppression.

In two dimensions: the gradient

- ▶ An image varies in two directions, so the derivative becomes a vector, the *gradient* $\nabla I = (\partial I / \partial x, \partial I / \partial y)$.
- ▶ Its direction points across the boundary and its magnitude says how sharp the boundary is. For edge detection we usually want the magnitude only, because an edge is an edge whatever its orientation:

$$f(I(x)) = |\nabla I(x)| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}.$$

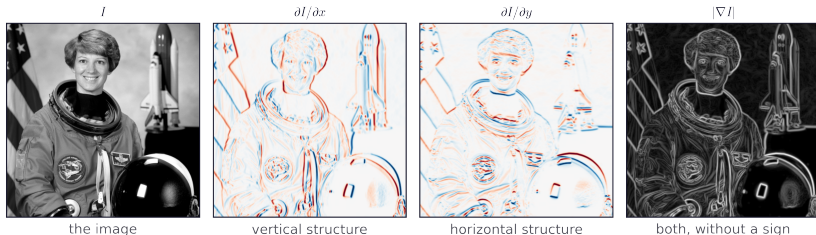


Figure 7: The image, its two partial derivatives, and the gradient magnitude. The two derivative images carry a sign, and the magnitude discards it.

Filterbanks: orientation and scale

- ▶ The gradient magnitude is one filter among many. A *filterbank* is a set of filters, each tuned to a different orientation, scale, or type of local structure.
- ▶ A common choice is the Gabor filter, a sinusoid multiplied by a Gaussian envelope,

$$g_{\theta,\lambda,\sigma}(x,y) = \exp\left(-\frac{x_{\theta}^2 + \gamma^2 y_{\theta}^2}{2\sigma^2}\right) \cos\left(\frac{2\pi x_{\theta}}{\lambda} + \phi\right),$$

where x_{θ}, y_{θ} are the coordinates rotated by θ . The orientation θ selects the direction the filter responds to, the wavelength λ and the envelope σ select the scale, and the phase ϕ selects between bar-like and edge-like structure.

- ▶ Different filters answer different questions about the same pixels.

Filterbanks: orientation and scale

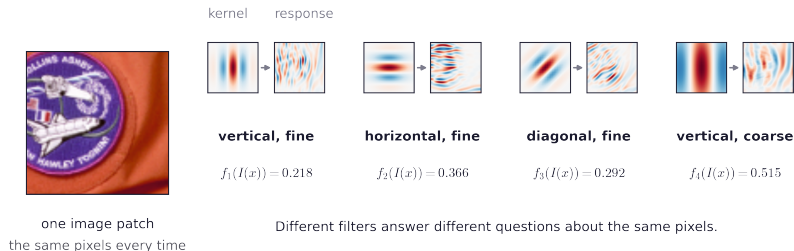


Figure 8: Four Gabor filters applied to one image patch. The kernel is shown beside the response it produces, and each filter returns one number for the patch.

- ▶ Two other cues are standard in the literature on statistical edge detection. The Laplacian of a Gaussian responds to intensity peaks and troughs rather than to steps. The Nitzberg operator is built from the eigenvalues of the smoothed outer product of the image gradient with itself, and therefore separates a single oriented structure from a corner or a texture. Konishi et al. use all three, at several scales.

One filter produces two distributions for pixels

- ▶ We assume an annotated dataset: a collection of images in which every pixel has been labelled by hand as edge or non-edge.
- ▶ Apply the filter everywhere. Each pixel then carries a pair $(f(I(x)), y(x))$. Split the responses into two pools according to the label and build a histogram of each pool. Formally the two distributions are estimated from

$$P(f(I) \mid y = +1) \quad \text{from} \quad \{f(I(x)) : y(x) = +1\},$$

$$P(f(I) \mid y = -1) \quad \text{from} \quad \{f(I(x)) : y(x) = -1\}.$$

- ▶ Every labelled pixel contributes exactly one response to exactly one of the two histograms. Each histogram is normalised by its own total, so each integrates to one. Each is a distribution over responses conditioned on the label.

One filter produces two distributions for pixels

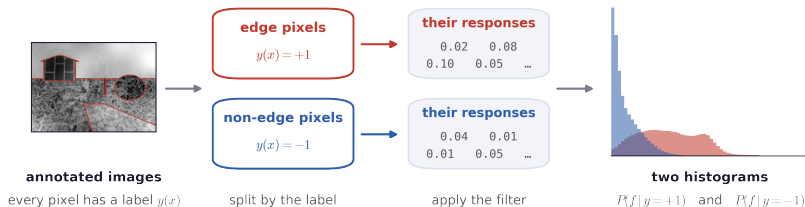


Figure 9: The two distributions are estimated by counting the responses of one filter on annotated data.

- ▶ The annotation enters the method only at this step. The decision rule that follows uses the two histograms and nothing else from the dataset.

Histogram estimation, stated precisely

- ▶ Divide the range of the response into B equally spaced bins $[b_{k-1}, b_k)$. Let N_+ be the number of labelled edge pixels and n_k^+ the number of them whose response falls in bin k . The estimate is

$$\hat{P}(f \in [b_{k-1}, b_k) \mid y = +1) = \frac{n_k^+}{N_+}, \quad \sum_{k=1}^B \frac{n_k^+}{N_+} = 1,$$

and similarly for $y = -1$ with N_- and n_k^- .

- ▶ The bin width controls a trade-off. Wide bins give a smooth estimate that blurs real structure; narrow bins follow the data closely at the price of noise, and an empty bin produces a log-likelihood ratio of $\pm\infty$. In practice the counts are floored at a small value before taking logarithms.
- ▶ The estimate converges to the true density as $N \rightarrow \infty$ with the bin width shrinking slowly enough that each bin still receives many samples.

Reading the two distributions

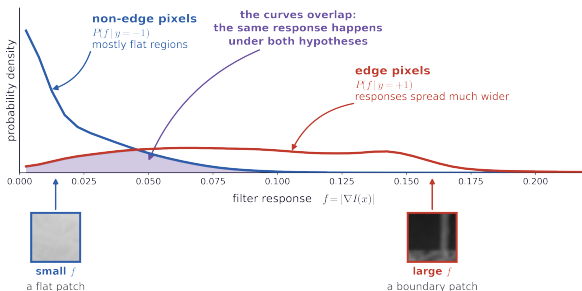


Figure 10: The two class-conditional distributions on one axis. The horizontal axis is the filter response, and the vertical axis is probability density.

- ▶ Read the figure as follows. Fix a response value on the horizontal axis. The height of the blue curve is how often that response occurs among non-edge pixels, and the height of the red curve is how often it occurs among edge pixels.
- ▶ The curves overlap, so the same response value occurs under both hypotheses. The decision has to weigh the two densities against each other, which is what makes edge detection a decision problem.

Why the non-edge curve piles up at zero

- ▶ Most pixels in a photograph lie inside a smooth region, so most derivative responses are close to zero. The non-edge distribution is therefore strongly peaked at the origin.
- ▶ This is a fact about *natural-image statistics*. The derivative of a photograph has a sharp peak at zero and long tails, and the derivative of white noise with the same pixel variance has a Gaussian shape with neither feature.
- ▶ A convenient model is the generalised Gaussian,
 $P(f) \propto \exp(-|f/s|^p)$ with p typically between 0.5 and 1, which reduces to the Laplace distribution at $p = 1$. Simoncelli and Olshausen, *Natural image statistics and neural representation* (2001), survey this literature.

Image statistics and the class prior

The heavy-tailed shape is a property of natural images. The class prior $P(y)$ is a separate quantity: it says how often a pixel is an edge.

Why the non-edge curve piles up at zero

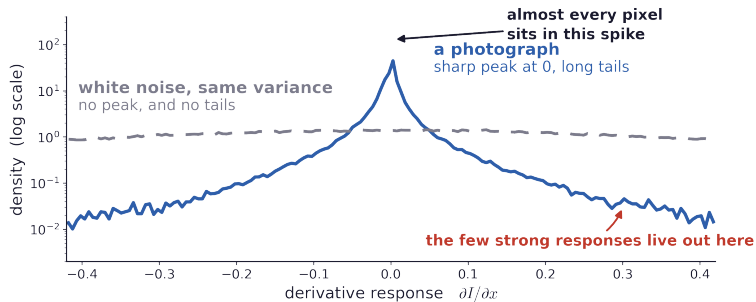


Figure 11: The horizontal derivative of a photograph, and of white noise with the same pixel variance, on a logarithmic density axis.

- The consequence for edge detection is that the two classes are unbalanced in shape as well as in frequency. Almost all of the probability mass of $P(f \mid y = -1)$ sits in a narrow spike, while $P(f \mid y = +1)$ is spread over a wide range.

Histograms assume nothing about the shape

- ▶ A histogram is a *non-parametric* estimate. It imposes no functional form and will represent whatever shape the data has, including shapes that no standard density would produce.
- ▶ The price is data. A parametric model such as a Gaussian or a generalised Gaussian needs only two or three numbers per class and can be fitted from far fewer samples, at the price of imposing a shape that may be wrong.
- ▶ For a single filter the histogram is affordable, and it is what Konishi et al. use. For many filters jointly the amount of data it needs becomes the binding constraint.

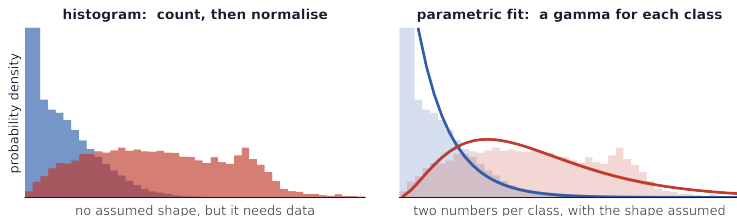


Figure 12: The same two classes estimated by a histogram and by a fitted gamma density.

Evaluating one response under both hypotheses

- ▶ Suppose the filter returns the value f^* at some pixel. Two questions can be asked of the two curves: how likely is f^* if this pixel is an edge, which is $P(f^* | y = +1)$, and how likely is it if the pixel is non-edge, which is $P(f^* | y = -1)$.
- ▶ The decision depends on which of the two densities is larger, and by how much. Both can be small at the same response, and the comparison between them is still informative.

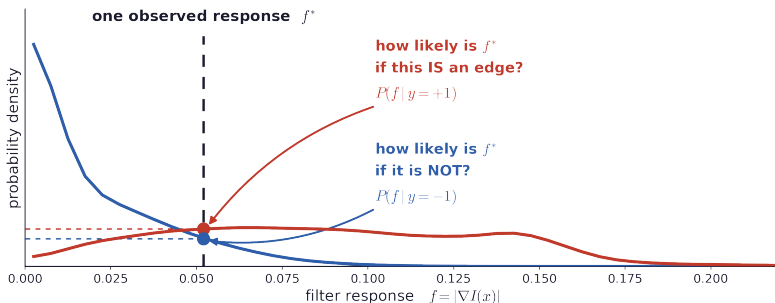


Figure 13: One observed response, read off both curves.

The likelihood ratio

- ▶ The comparison is the *likelihood ratio*, and it is conventional to take its logarithm so that the two directions are symmetric and so that independent evidence adds rather than multiplies:

$$\log \frac{P(f(I(x)) \mid y = +1)}{P(f(I(x)) \mid y = -1)}.$$

- ▶ The ratio takes large positive values where the response is more typical of edges, and large negative values where it is more typical of background. The sign says which hypothesis the evidence favours, and the magnitude says by how much.

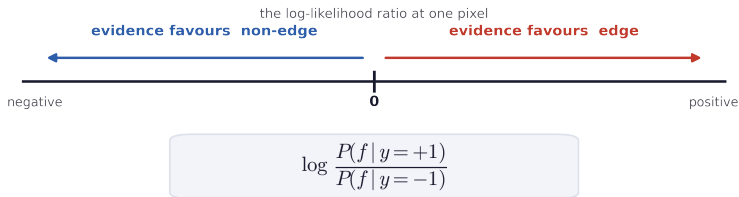
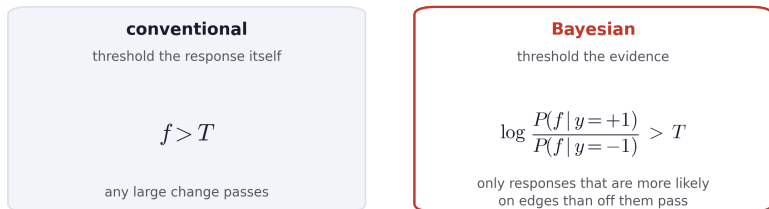


Figure 14: The log-likelihood ratio at one pixel, on a signed axis.

The Bayesian test thresholds the likelihood ratio

- ▶ A conventional detector thresholds the response itself, $f > T$. A Bayesian detector thresholds the evidence the response carries.
- ▶ The two tests agree whenever the likelihood ratio increases monotonically with f , which happens for instance if $P(f | y = +1)$ is a shifted copy of $P(f | y = -1)$. For the distributions measured on real images the ratio rises and falls, and the two tests then differ.



The threshold moves from the raw number to the evidence it carries.

Figure 15: Thresholding the raw response, and thresholding the evidence.

The threshold comes from the prior and the loss

- ▶ Testing whether the log-likelihood ratio exceeds zero is a special case. It assumes that edge and non-edge are equally likely, and that the two kinds of error cost the same. Edge detection breaks both assumptions: edges are rare, and a missed boundary costs more than a spurious one.
- ▶ Write c_{FP} for the cost of declaring an edge where there is none, and c_{FN} for the cost of missing one, with correct decisions costing nothing. By Bayes decision theory we declare an edge when the expected cost of doing so is smaller:

$$P(y = -1 | f) c_{FP} < P(y = +1 | f) c_{FN}.$$

- ▶ Substituting $P(y | f) = P(f | y)P(y)/P(f)$ and cancelling the common $P(f)$,

$$P(f | y = -1) P(y = -1) c_{FP} < P(f | y = +1) P(y = +1) c_{FN}.$$

The threshold comes from the prior and the loss

- ▶ Dividing and taking logarithms gives the rule in its final form:

$$\log \frac{P(f \mid y = +1)}{P(f \mid y = -1)} > T, \quad T = \underbrace{\log \frac{P(y = -1)}{P(y = +1)}}_{\text{the prior term}} + \underbrace{\log \frac{c_{FP}}{c_{FN}}}_{\text{the loss term}}$$

- ▶ **The prior term.** Most pixels are not edges, so a pixel has to be unusual before we call it one. This raises T . With roughly 6% of pixels labelled as edges the term is $\log(0.94/0.06)$.
- ▶ **The loss term.** Missing a boundary usually costs more than reporting a spurious one, because later stages can use context to remove false edges and cannot recover a boundary that was never detected. This lowers T . Treating a miss as ten times worse gives $\log(1/10)$.
- ▶ The two terms pull in opposite directions, and with these numbers they very nearly cancel.

The decision rule in compact form

- It is convenient to write the rule as a maximisation over the two labels rather than as an inequality. Define

$$\alpha_T(x) = \arg \max_{y \in \{\pm 1\}} y \left\{ \log \frac{P(f(I(x)) \mid y = +1)}{P(f(I(x)) \mid y = -1)} - T \right\}.$$

- To see that this is the same rule, let Λ denote the bracket. For $y = +1$ the objective is Λ , and for $y = -1$ it is $-\Lambda$. The maximum is attained at $y = +1$ exactly when $\Lambda > 0$, which is exactly the test above.
- Setting $T = 0$ recovers the rule that assumes a uniform prior over the two classes and equal costs.

Moving the threshold

- ▶ A detector makes two kinds of mistake. A *false positive* is a non-edge pixel declared an edge, and a *false negative* is an edge pixel declared non-edge.
- ▶ Lowering T detects more edges, and more texture with them: false positives rise and false negatives fall. Raising T keeps only the strongest evidence: false positives fall and real boundaries are lost.
- ▶ To classify every pixel as non-edge the threshold must be sent to $T \rightarrow +\infty$, since the rule declares an edge when the ratio *exceeds* T .

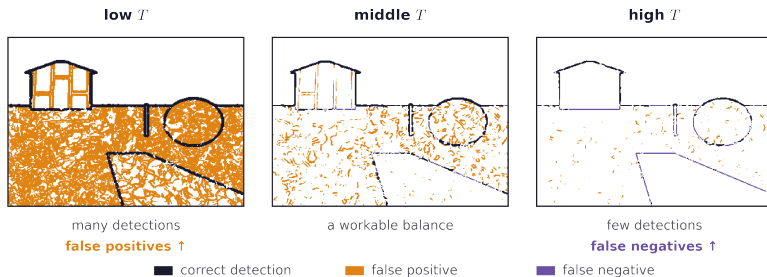


Figure 16: The same image at three thresholds, with correct detections, false positives and false negatives marked separately.

How performance is measured

- ▶ Three measures are used in this literature, and Konishi et al. report all three.
- ▶ The **Bayes risk** is the expected loss of the best rule. With 0–1 loss it is the total error probability,
$$R = P(y=+1) P(\text{miss}) + P(y=-1) P(\text{false alarm}).$$
- ▶ The **receiver operating characteristic**, or ROC curve, plots the detection rate against the false positive rate as T is swept. The area under it summarises the detector without committing to a threshold.
- ▶ The **Chernoff information** measures how far apart the two class-conditional distributions are,

$$C = - \min_{0 \leq \lambda \leq 1} \log \int P(f \mid y = +1)^\lambda P(f \mid y = -1)^{1-\lambda} df,$$

and gives the asymptotic exponential rate at which the error falls as independent measurements are accumulated. It depends on the two distributions alone, and is unaffected by the prior and the loss.

The full pipeline on one image

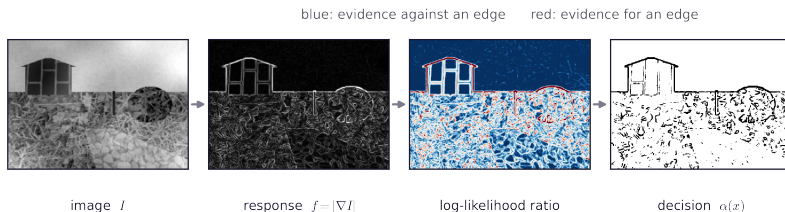


Figure 17: The third panel is the log-likelihood ratio computed at every pixel from the two learned distributions. The fourth is that map thresholded at T .

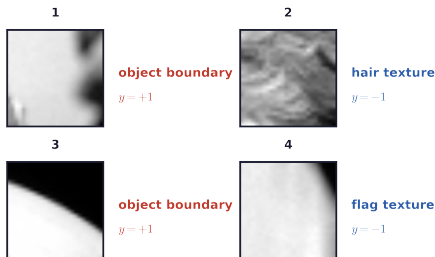
- The four stages are the image, the filter response, the evidence, and the decision. Each extension that follows replaces one stage: a vector of responses in place of one number, a joint distribution in place of two histograms, or a different threshold.

Local evidence is ambiguous

- ▶ Even a human cannot label a pixel reliably from a small neighbourhood alone. Shown the whole image, every boundary is obvious; shown a few pixels in isolation, a true boundary and a piece of texture look alike.
- ▶ Two remedies exist. The first is to measure more at the same location, by applying a bank of filters. The second is to use information from elsewhere in the image, either spatial context or knowledge of the objects present.



the whole image: every boundary is obvious



the same four places, seen alone

Figure 18: Four small windows. In the whole image each one is easy to name, and seen alone two boundaries and two textures look alike.

Many filters at one pixel

- ▶ Apply a filterbank $\{f_i(\cdot) \mid i = 1, \dots, M\}$ at the same pixel. The observation is now a vector of responses rather than a single number, and the decision rule becomes

$$\alpha_T(x) = \arg \max_{y \in \{\pm 1\}} y \left\{ \log \frac{P(f_1, \dots, f_M \mid y = +1)}{P(f_1, \dots, f_M \mid y = -1)} - T \right\}.$$

- ▶ The rule keeps its structure. The object that must be estimated becomes a *joint* distribution over M responses, conditioned on the label.

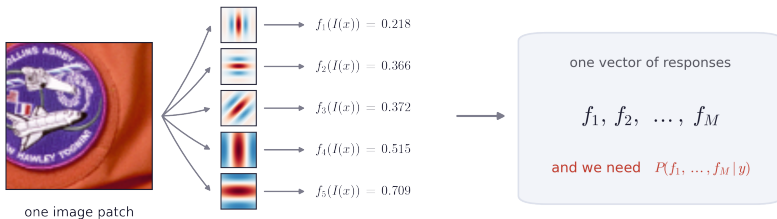


Figure 19: One patch, five filters, one response vector.

Learning the joint distribution is hard

- ▶ Suppose each response is quantised into B bins. A one-dimensional histogram has B cells, a two-dimensional histogram has B^2 , and the joint histogram over M filters has B^M .
- ▶ With $B = 10$ and only $M = 6$ filters there are already 10^6 cells. The number of labelled pixels available stays fixed as M grows, so almost every cell is empty and the estimate is dominated by sampling noise.
- ▶ There is a second objection, independent of the data. A filled-in B^M table gives the answer and says nothing about *which* cue produced it, so it offers no insight and no way to diagnose failures.

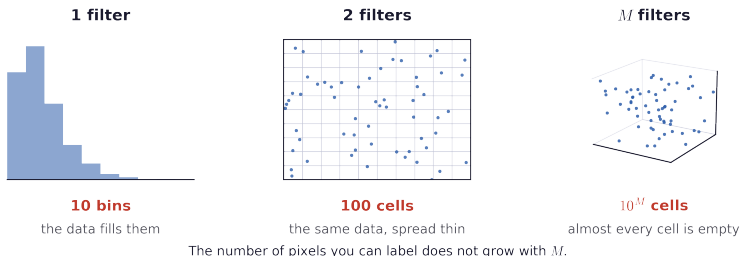
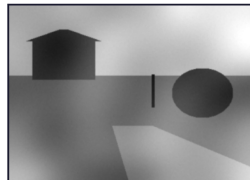


Figure 20: The same fixed number of samples, spread over one, two and many dimensions.

- ▶ Konishi, Yuille, Coughlan and Zhu made the first attempt to detect edges from annotated datasets and probabilities, and to evaluate the result quantitatively.
- ▶ **Sowerby**: one hundred pre-segmented colour images of English outdoor scenes, all containing a road or track, with a great deal of vegetation. **South Florida**: fifty greyscale images, largely indoor, with very little background texture.
- ▶ Roughly 6% of pixels are labelled as edges in both.



countryside: vegetation everywhere



indoor: quiet backgrounds

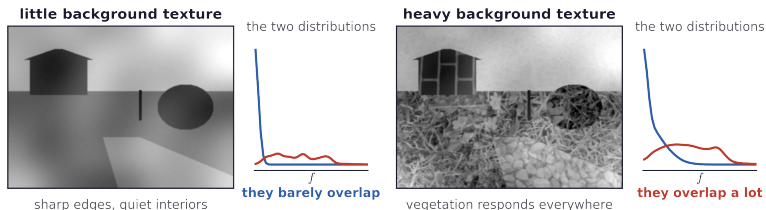
Figure 21: Countryside with vegetation everywhere, and quiet indoor scenes.

- ▶ The cues were the gradient magnitude, the Laplacian and the Nitzberg operator, each at several scales and on several colour bands, together with oriented filterbanks built from Gabor and Hilbert transform pairs.
- ▶ The distributions $P(f \mid y = +1)$ and $P(f \mid y = -1)$ were estimated as histograms for each cue, from the hand-labelled edge maps supplied with the datasets.
- ▶ Performance was reported with all three measures: Bayes risk, ROC area and Chernoff information.

What Konishi et al. found

- ▶ **On South Florida the learned detector matched a conventional one.** The statistical edge detector achieved a Bayes risk of 0.0350, and their implementation of the Canny detector achieved 0.0352. Where the background carries no texture, thresholding a single derivative cue is already close to optimal, and cue combination has little left to gain.
- ▶ **On Sowerby the learned detector clearly won.** Canny gave a Bayes risk of 0.211; statistical edge detection gave 0.182 without grouping and 0.178 once edges were grouped. Vegetation produces many strong derivative responses at pixels that are not object boundaries, and a single cue lumps them together with real edges.
- ▶ Multi-scale processing helped most on Sowerby, and the Nitzberg operator was the most effective single cue for discriminating edges from texture.

Why the two datasets behave differently



A single derivative cue is enough on the left, and fails on the right.

Figure 22: Two scenes with identical region boundaries and identical region means, differing only in the texture inside the regions, and the two class-conditional distributions each produces.

- ▶ The boundaries are the same in both, so $P(f | y = +1)$ is much the same. The interiors differ, so $P(f | y = -1)$ differs a great deal, and with it the overlap between the two curves. The overlap is what determines the Bayes risk.

Independent cues: the evidence adds

- Suppose the filter responses were statistically independent given the label, for each y :

$$P(f_1, \dots, f_M | y) = \prod_{i=1}^M P(f_i | y).$$

- Then the log-likelihood ratio factorises into a sum,

$$\log \frac{P(f_1, \dots, f_M | y = +1)}{P(f_1, \dots, f_M | y = -1)} = \sum_{i=1}^M \log \frac{P(f_i | y = +1)}{P(f_i | y = -1)},$$

and the decision rule becomes

$$\alpha_T(x) = \arg \max_{y \in \{\pm 1\}} y \left\{ \sum_{i=1}^M \log \frac{P(f_i | y = +1)}{P(f_i | y = -1)} - T \right\}.$$

Independent cues: the evidence adds

- ▶ **The data requirement collapses.** Only M one-dimensional distributions have to be learned. The requirement grows linearly in M rather than exponentially, so a filterbank of twenty filters is as cheap to estimate as twenty single filters.
- ▶ **The decision becomes interpretable.** Each filter contributes a signed number, a vote for or against an edge, and the decision is the sign of the sum offset by T . Reading off which filters carried the decision is immediate.

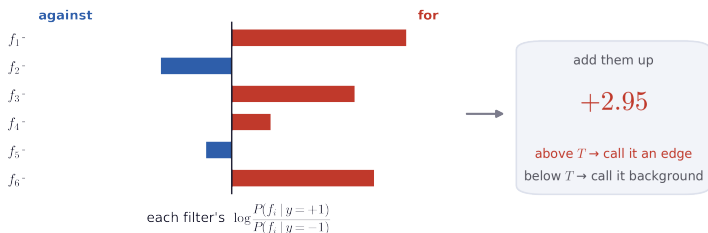


Figure 23: Each filter casts a signed vote, and the decision is the sign of their sum, offset by the threshold T .

The filters are correlated

- ▶ Statistical independence is rare in practice. The response of each filter depends on the total brightness or contrast of the image patch, so all of them respond more strongly to a strong edge than to a weak one, and their responses rise and fall together.
- ▶ The figure shows this directly: a fine-scale and a coarse-scale derivative measured at the same pixels are strongly correlated.

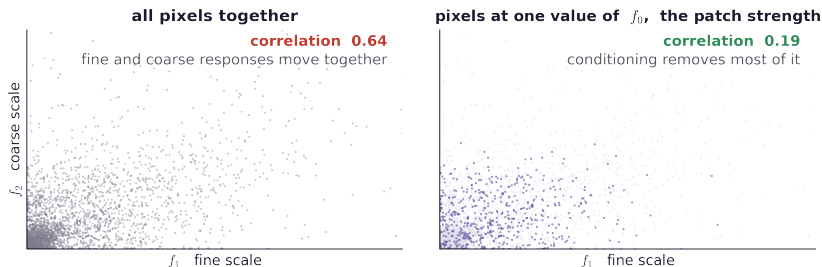


Figure 24: Fine-scale against coarse-scale response. Left: all pixels together. Right: only pixels at one value of the patch strength f_0 .

Conditional independence given the patch strength

- ▶ This suggests a weaker condition. Add one further filter $f_0(I(x))$ which measures the overall brightness, or strength, of the patch. It is then possible that the other filters are independent *conditioned on* the value of f_0 :

$$\begin{aligned} &P(\{f_i(I(x))\}, f_0(I(x)) \mid y) \\ &= P(f_0(I(x)) \mid y) \prod_{i=1}^M P(f_i(I(x)) \mid f_0(I(x)), y). \end{aligned}$$

- ▶ Only two families of distributions then have to be represented: $P(f_i \mid f_0, y)$ and $P(f_0 \mid y)$. The right-hand panel of the previous figure shows the effect: conditioning on f_0 removes most of the correlation.

The decision rule with conditioning

- ▶ Substituting the factorisation into the log-likelihood ratio gives a rule that is still a sum of per-filter votes, with one extra term for the conditioning filter. Write f_i for $f_i(I(x))$:

$$\alpha_T(x) = \arg \max_{y \in \{\pm 1\}} y \left\{ \log \frac{P(f_0 \mid y = +1)}{P(f_0 \mid y = -1)} + \sum_{i=1}^M \log \frac{P(f_i \mid f_0, y = +1)}{P(f_i \mid f_0, y = -1)} - T \right\}$$

- ▶ It has been argued that methods of this type can be implemented by neurons and may be responsible for edge detection in biological vision: Ramachandra and Mel, *Computing local edge probability in natural scenes from a population of oriented simple cells* (2013). The argument is general and holds whatever the filters $f_i(\cdot)$ are, linear or otherwise. Morrone and Burr, *Feature detection in human vision: a phase-dependent energy model* (1988), propose that edge detection uses the energy model of complex cells.

The same machinery, more classes

- ▶ The same argument works for any number of classes. It applies in particular to *semantic segmentation*: giving every location in the image a class label, the way every pixel has been given an edge label.
- ▶ Take $y \in \mathcal{Y}$ with, for example, $\mathcal{Y} = \{\text{sky, vegetation, water, road, other}\}$, choose a set of filters $\{f_i(I(x))\}$ sensitive to texture and to colour, and learn one distribution $P(\{f_i(I(x))\} \mid y)$ per class exactly as before.

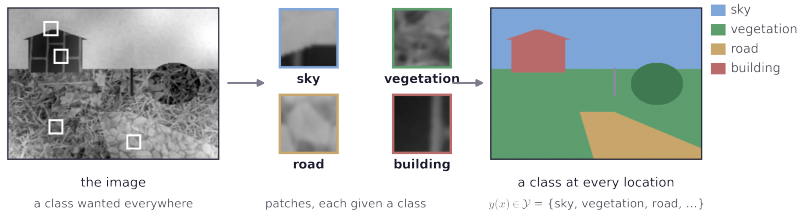


Figure 25: A class label is wanted at every location. Local patches are classified one at a time.

The multiclass rule

- ▶ Bayes decision theory gives the rule directly. Minimising the expected loss over $|\mathcal{Y}|$ classes yields

$$\alpha(I(x)) = \arg \max_{y \in \mathcal{Y}} P(\{f_i(I(x))\} | y) T_y,$$

where T_y is one weight per class rather than a single number.

- ▶ T_y carries what the prior and the loss carried in the binary case: how common the class is, and what it costs to get it wrong. It is derived from decision theory in the same way as T was above.
- ▶ With two classes and equal weights this reduces to the likelihood ratio test, since $P(f | +1)T_{+1} > P(f | -1)T_{-1}$ is the same inequality as $\log P(f | +1)/P(f | -1) > \log T_{-1}/T_{+1}$. The threshold of the binary rule is $T = \log T_{-1}/T_{+1}$.

From local classes to scene parsing

- ▶ Konishi and Yuille (2000) applied exactly this rule, with the classes edge, vegetation, air, road, building and other, using intensity, the gradient magnitude, the Nitzberg operator and the Laplacian of a Gaussian at several scales, on the Sowerby database and a set of San Francisco street scenes.
- ▶ Accuracy above 90% was obtained for the spatially homogeneous classes, road, vegetation and air. Edge and building were classified poorly from local filters alone, the same ambiguity that a single edge cue runs into.
- ▶ Modern work poses a much richer version of the task. Mottaghi et al. (2014) annotated 10,103 images from PASCAL VOC 2010 with 540 categories of object and stuff; the 59 most frequent classes cover 87.2% of pixels. Their point is that context substantially improves recognition at all object scales.

From local classes to scene parsing

local cues, one pixel at a time

the idea in Konishi & Yuille, 2000

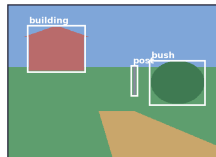


large uniform regions come out well
thin structure and object detail do not

richer models,
→
more context

every region and object named

the setting of Mottaghi et al., 2014



hundreds of classes, objects and stuff
context decides what the local cue cannot

Figure 26: The classes a local-cue classifier recovers, and the labelling that modern scene parsing aims at.

- ▶ Local cues alone already predict the spatially homogeneous classes, sky, grass and water. Complete scene understanding, with hundreds of object categories and their parts, needs richer models and spatial context.

Bayes versus direct prediction

- ▶ Konishi et al. formulated edge detection as Bayesian inference: learn the generative distributions $P(f(I) | y = +1)$ and $P(f(I) | y = -1)$ and a prior $P(y)$, then invert them with Bayes' rule.
- ▶ Many researchers learn $P(y | f(I))$ directly from labelled examples, without modelling how a class produces responses. Deep networks are the most popular methods of this kind.
- ▶ Both routes end at a decision. They differ in what is modelled on the way there, and so in what can be reused when circumstances change.

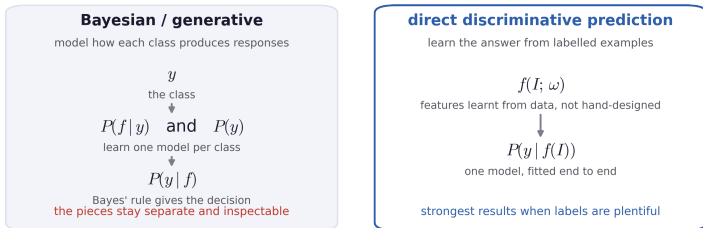


Figure 27: Two routes to the same posterior $P(y | f)$.

What deep networks changed

- ▶ In the statistical edge detectors the filters were chosen by hand: derivatives, Laplacians, the Nitzberg operator, Gabor functions. Only the distributions over their responses were learned.
- ▶ A deep network learns the features as well. The response becomes $f(I; \omega)$, a hierarchical parameterised function whose weights ω are fitted along with the classifier. A loss function is specified, and it can be chosen to reward correct detections at several levels of the network, reflecting the fact that image cues occur at multiple scales.
- ▶ Direct prediction is the current state of the art on annotated benchmark datasets. Xie and Tu, *Holistically-Nested Edge Detection* (2015), is the standard reference for edge detection.
- ▶ The Bayesian formulation keeps the model in named pieces: one distribution for edges, one for the background, a prior and a loss. When something about the world changes, it is often possible to say which piece changed and re-estimate only that piece.

Domain transfer is a problem for vision algorithms

- ▶ The statistical properties of images differ greatly between domains. An algorithm trained on one domain may fail completely on another, because it has been fitted to statistics that no longer hold.
- ▶ The two datasets of this lecture are a clean example. Sowerby contains sharp edges at the boundaries of roads, buildings and houses, and a great deal of texture in the background regions, mostly vegetation. South Florida consists of indoor images where the edges are sharp and the background carries almost no texture.
- ▶ An edge detector trained on South Florida will perform badly on Sowerby, because it will report edges throughout the vegetation.

What actually changes between domains

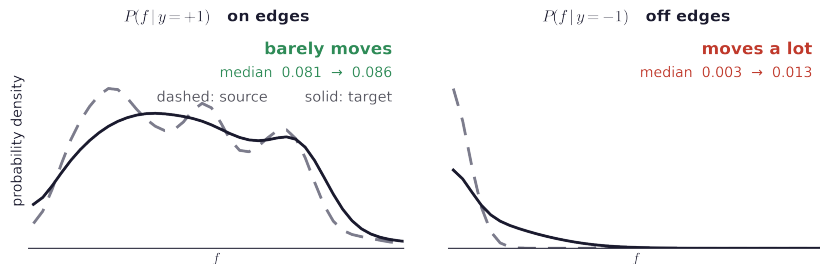


Figure 28: The on-edge and off-edge response distributions for two domains that share the same boundaries. Dashed: source. Solid: target.

- **Observation I.** $P(f(I) | y = +1)$ is similar between the two datasets, because the edges have similar properties in each. A boundary of a given contrast produces much the same response wherever the picture was taken.
- **Observation II.** $P(f(I) | y = -1)$ changes a great deal. It is the distribution of the background, and the background is what distinguishes the two domains.

Domain transfer using the factorisation

- ▶ These two observations together give a procedure that needs annotation in one domain only.
- ▶ **Step 1.** Learn $P(f(I) \mid y = +1)$ from the labelled source domain and keep it, by Observation 1.
- ▶ **Step 2.** Re-estimate $P(f(I) \mid y = -1)$ on the target domain. This step needs no annotation. Since 90 to 95% of the pixels in an image are not edges, the histogram of responses over *all* target pixels is already a good approximation to the off-edge distribution.
- ▶ **Step 3.** Form the likelihood ratio from the source on-edge model and the target off-edge model, and apply the same decision rule.

$$\log \frac{P_{\text{source}}(f \mid y = +1)}{P_{\text{target}}(f \mid y = -1)} > T$$

Domain transfer using the factorisation

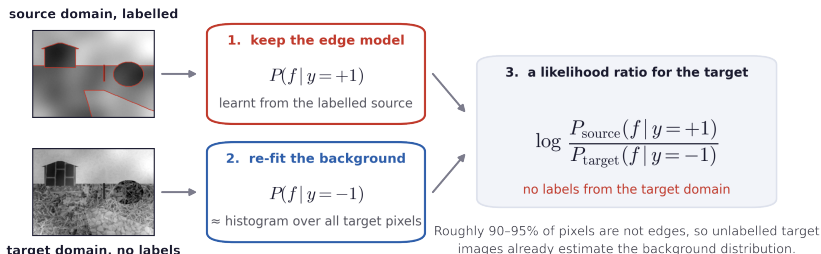


Figure 29: A detector for the target domain, built from labelled source images and unlabelled target images.

- ▶ Konishi et al. report that this works. Adapting the distributions learned on one dataset to the other cost very little: the ROC area fell from 0.877 to 0.867, and the Chernoff information from 0.336 to 0.322, against distributions learned on the target itself.
- ▶ Direct prediction finds this transfer hard unless training data is available for both domains, because the quantity it models, $P(y | f)$, mixes together the part that transfers and the part that does not.

Summary

- ▶ Edge detection was written as a decision at every pixel, edge or non-edge, which is the problem Bayes decision theory solves.
- ▶ The observation is a filter response. The two class-conditional distributions were learned by counting responses on annotated images.
- ▶ The decision compares the two distributions at the observed response and thresholds that comparison. The threshold comes from the prior and the loss.
- ▶ One filter is often ambiguous. Several filters can be combined, and under conditional independence the evidence adds, which keeps both the data requirement and the interpretation manageable.
- ▶ The same rule extends to many classes, and gives semantic segmentation.
- ▶ Because the model is written in named pieces, the background distribution can be re-estimated for a new domain while the edge distribution is kept.
- ▶ Local evidence stays ambiguous. Spatial context and object knowledge are the two remaining ways out.

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